

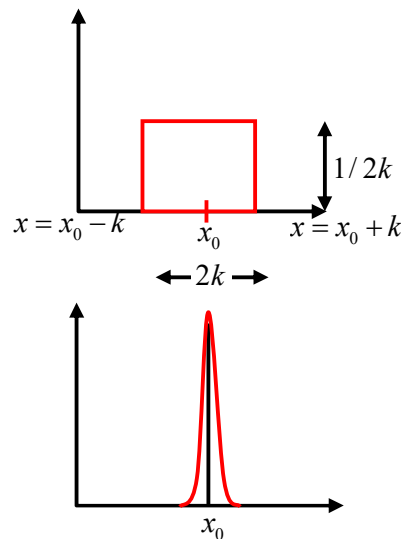
Fourier Transformation

2. Dirac Delta Function

$$\Rightarrow A = \frac{1}{2k} \times 2k = 1$$

$$\Rightarrow 2k \rightarrow 0, \frac{1}{2k} \rightarrow \infty$$

$$\text{Then } f(x) = \delta(x - x_0)$$

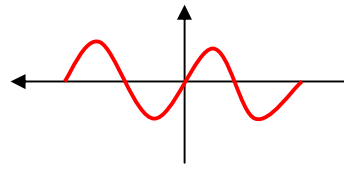


Properties:

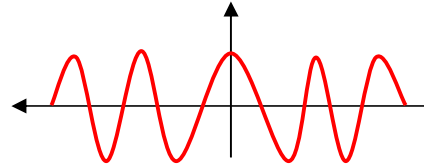
- The area under the curve: $\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \delta(x-x_0) dx = 1$ (Area is unity)
- $\int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx = f(x_0) \int_{-\infty}^{\infty} \delta(x-x_0) dx = f(x_0), \quad x \in]-\infty, \infty[$
- $\int_{-\infty}^{\infty} e^{ix} \delta(x-x_0) dx = e^{ix_0}$

In certain limits, many functions behave like the Dirac delta function

$$\bullet \int_{-\infty}^{\infty} e^{-j2\pi(f-f_0)t} dt = \delta(f-f_0)$$

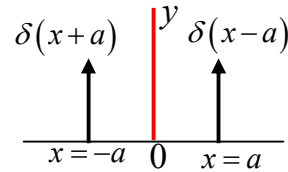


$$\bullet \int_{-\infty}^{\infty} e^{-j2\pi(f+f_0)t} dt = \delta(f+f_0)$$



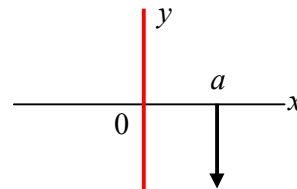
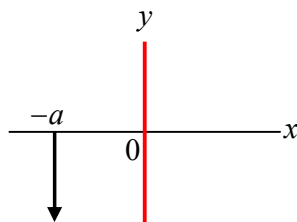
Integral become zero of sine and cosine except at $f = f_0$.

$$\bullet \delta(x+a) + \delta(x-a) \text{ if } a > 0$$



$$\bullet -\delta(x+a) \text{ if } a > 0$$

$$-\delta(x-a) \text{ if } a > 0$$



$$\bullet \int_{-\infty}^{\infty} \delta(ax) dx, \text{ Consider } ax = y, \quad adx = dy$$

$$\Rightarrow \frac{1}{a} \int_{-\infty}^{\infty} \delta(y) dy = \frac{1}{a}$$

It has importance when its area is finite and height is ∞ .

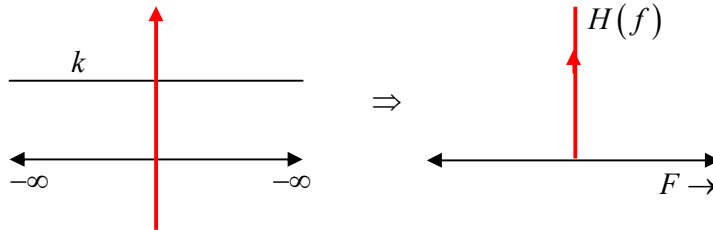
Example: If $h(t) = k$, Then find its Fourier Transform.

Solution: $H(f) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft} dt = k \int_{-\infty}^{\infty} e^{0t} e^{-j2\pi ft} dt$

$$= k \int_{-\infty}^{\infty} e^{0j2\pi ft} e^{-j2\pi ft} dt$$

$$= k \int_{-\infty}^{\infty} e^{-j2\pi(f-0)t} dt$$

$$H(f) = k\delta(f-0) = k\delta(f)$$



Example: If $h(t) = A \sin 2\pi f_0 t = \frac{A}{2j} [e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}]$, where f_0 is the frequency of signal and

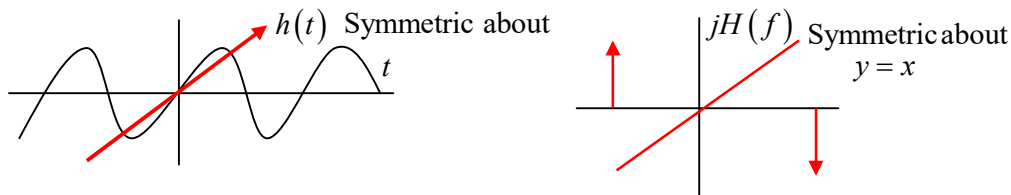
f is the frequency of Fourier signal. Then find its Fourier transform.

Solution: $H(f) = \int_{-\infty}^{\infty} \frac{A}{2j} [e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}] e^{-j2\pi ft} dt$

$$= \frac{A}{2j} \int_{-\infty}^{\infty} e^{-j2\pi(f-f_0)t} dt - \frac{A}{2j} \int_{-\infty}^{\infty} e^{-j2\pi(f+f_0)t} dt$$

$$H(f) = \frac{A}{2j} [\delta(f-f_0) - \delta(f+f_0)] = -j \frac{A}{2} [\delta(f+f_0) - \delta(f-f_0)]$$

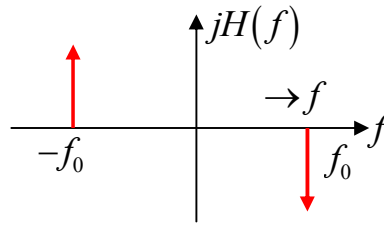
$\Rightarrow$ Symmetric about x-axis



(i) If a function is odd then the Fourier transformation will be imaginary.

(ii) If a function is even then the Fourier transformation will be real.

(iii) If a function is mixed then the Fourier transformation will be complex.



Sampling should be greater than (f_0)

⇒ The analogy between Fourier series with Fourier transformation:

$$H(f) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft} dt$$

If $h(t) \rightarrow$ even and $e^{-j2\pi ft} = \cos 2\pi ft - j \sin 2\pi ft$

$$H(f) = \int_{-\infty}^{\infty} \left\{ \underbrace{h(t)(\cos 2\pi ft)}_{\text{even integral} \neq 0} - \underbrace{j(\sin 2\pi ft) \times h(t)}_{\text{odd integral} = 0} \right\} dt$$

$$H(f) = \text{Real}$$

Similarly

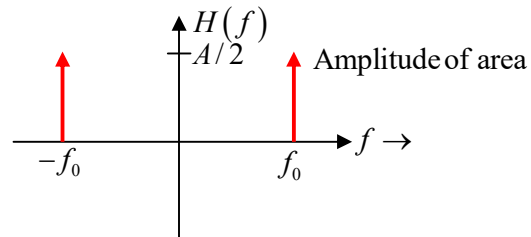
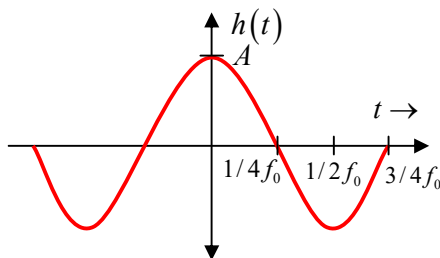
If $h(t) \rightarrow$ odd then $H(f) = \text{imaginary}$

Example: If $h(t) = A \cos 2\pi f_0 t$, where f_0 is the frequency of the signal and f is the frequency of Fourier signal. Then find its Fourier transform.

Solution: $H(f) = \int_{-\infty}^{\infty} \frac{A}{2} [e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}] e^{-j2\pi ft} dt$

$$= \frac{A}{2} \int_{-\infty}^{\infty} e^{-j2\pi(f-f_0)t} dt + \frac{A}{2} \int_{-\infty}^{\infty} e^{-j2\pi(f+f_0)t} dt$$

$$H(f) = \frac{A}{2} [\delta(f-f_0) + \delta(f+f_0)]$$



Time Domain

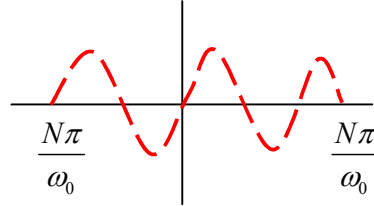
⇒ Symmetric about y -axis (even)

Frequency Domain

⇒ Symmetric about y -axis (even)

Example: A sinusoidal pulse is clipped and the resultant pulse is

$$f(t) = \begin{cases} \sin \omega_0 t & |t| < \frac{N\pi}{\omega_0} \\ 0 & |t| > \frac{N\pi}{\omega_0} \end{cases}$$



Then find its Fourier transform.

Solution:
$$F(\omega) = \int_{-N\pi/\omega_0}^{N\pi/\omega_0} \sin \omega_0 t e^{-j\omega t} dt$$

Now by
$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$= \left[\frac{e^{-j\omega t}}{(-j\omega)^2 + \omega_0^2} (-j\omega \sin \omega_0 t - \omega_0 \cos \omega_0 t) \right]_{-N\pi/\omega_0}^{N\pi/\omega_0}$$

$$= \left[\frac{e^{-j\omega N\pi/\omega_0}}{\omega_0^2 - \omega^2} \right] [-j\omega \sin N\pi - \omega_0 \cos N\pi] - \frac{e^{j\omega N\pi/\omega_0}}{\omega_0^2 - \omega^2} [-j\omega \sin N\pi - \omega_0 \cos N\pi]$$

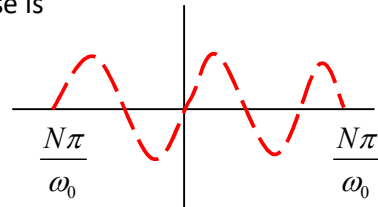
$$= \frac{j\omega \sin N\pi}{\omega_0^2 - \omega^2} [e^{j\omega N\pi/\omega_0} - e^{-j\omega N\pi/\omega_0}] + \frac{\omega_0 \cos N\pi}{\omega_0^2 - \omega^2} [e^{j\omega N\pi/\omega_0} - e^{-j\omega N\pi/\omega_0}]$$

$$= 2j \frac{\omega_0 \cos N\pi}{\omega_0^2 - \omega^2} \left[\sin \left(\frac{\omega N\pi}{\omega_0} \right) \right]$$

$$F(\omega) = \frac{\omega_0 (-1)^N}{\omega_0^2 - \omega^2} 2j \sin \frac{\omega N\pi}{\omega_0}$$

Example: A sinusoidal pulse is clipped and the resultant pulse is

$$f(t) = \begin{cases} \sin \omega_0 t & |t| < \frac{N\pi}{\omega_0} \\ 0 & |t| > \frac{N\pi}{\omega_0} \end{cases}$$



Then find its Fourier transform by a unit step function.

Solution:
$$f(t) = \left\{ u \left(t + \frac{N\pi}{\omega_0} \right) - u \left(t - \frac{N\pi}{\omega_0} \right) \right\} \sin \omega_0 t$$

$$\begin{aligned}
 F(\omega) &= \int_{-N\pi/\omega_0}^{N\pi/\omega_0} \sin \omega_0 t e^{-j\omega t} dt = \int_{-N\pi/\omega_0}^{N\pi/\omega_0} \sin \omega_0 t [\cos(\omega t) - j \sin(\omega t)] dt \\
 &= \frac{-j}{2} \int_{-N\pi/\omega_0}^{N\pi/\omega_0} 2 \sin \omega_0 t \sin \omega t dt = \frac{-j}{2} \int_{-N\pi/\omega_0}^{N\pi/\omega_0} [\cos(\omega - \omega_0)t - \cos(\omega + \omega_0)t] dt \\
 &= \frac{-j}{2} \left[\frac{\sin(\omega - \omega_0)t}{\omega - \omega_0} - \frac{\sin(\omega + \omega_0)t}{\omega + \omega_0} \right]_{-N\pi/\omega_0}^{N\pi/\omega_0} \\
 &= \frac{-j}{2} \left[\frac{\sin(\omega - \omega_0) \frac{N\pi}{\omega_0}}{\omega - \omega_0} - \frac{\sin(\omega + \omega_0) \frac{N\pi}{\omega_0}}{\omega + \omega_0} + \frac{\sin(\omega - \omega_0) \frac{N\pi}{\omega_0}}{\omega - \omega_0} - \frac{\sin(\omega + \omega_0) \frac{N\pi}{\omega_0}}{\omega + \omega_0} \right] \\
 &= j \left[\frac{\sin(\omega + \omega_0) N\pi / \omega_0}{\omega + \omega_0} - \frac{\sin(\omega - \omega_0) N\pi / \omega_0}{\omega - \omega_0} \right] \\
 &= j \left[\frac{1}{\omega + \omega_0} \left(\sin \frac{\omega N\pi}{\omega_0} (-1)^N \right) - \frac{1}{\omega - \omega_0} \left(\sin \frac{\omega N\pi}{\omega_0} (-1)^N \right) \right] \\
 &= j (-1)^N \sin \frac{\omega N\pi}{\omega_0} \left[\frac{1}{\omega + \omega_0} - \frac{1}{\omega - \omega_0} \right] = j (-1)^N \sin \frac{N\pi\omega}{\omega_0} \left[\frac{-2\omega_0}{\omega^2 - \omega_0^2} \right]
 \end{aligned}$$

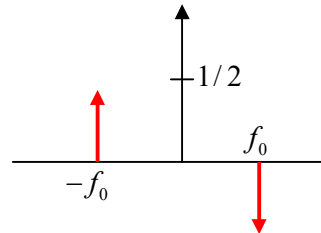
Example: Consider a sinusoidal waveform of amplitude 1 volt and the frequency f_0 starting from any arbitrary initial time. The waveform is sampled at intervals of $\frac{1}{2f_0}$. If the corresponding Fourier spectrum peaks at a frequency $\bar{f}$ and amplitude $\bar{A}$. Then find its Fourier transform.

Solution: $h(t) = 1 \cdot \sin 2\pi f_0 t$

$$H(f) = \frac{j}{2} [\delta(f - f_0) - \delta(f + f_0)]$$

$$\bar{A} = \frac{1}{2}, \bar{f} = f_0, -f_0$$

$$\Rightarrow \bar{f} = f_0, \quad 0 < \bar{A} < 1, \quad \bar{f} = -f_0, \quad -1 < \bar{A} < 0$$



Example: The Fourier transform of the derivative of the delta function namely $\delta'(x)$ is proportional to what?

Solution: $\delta'(x) = (-ik) \propto ik$

Example: The Fourier transform of $f(x)$ is $\tilde{f}(k) = \int_{-\infty}^{\infty} e^{ikx} f(x) dx$. Then find $\tilde{f}(k)$ for the following functions $f(x)$.

(a) $f(x) = \alpha \delta(x) + \beta \delta'(x) + \gamma \delta''(x)$

(b) $f(x) = \frac{1}{x^2 + 2}$

(c) $f(x) = e^{-|x|}$

(d) $f(x) = e^{a|x|} \sin bx$

(e) $f(x) = e^{-a|x|} \cos bx$

(f) $f(x) = \frac{[u(x+a) - u(x-\infty)]}{2a}$

Solution: (a) $\tilde{f}(k) = \int_{-\infty}^{\infty} \alpha \delta(x) e^{ikx} dx + \beta \int_{-\infty}^{\infty} \delta'(x) e^{ikx} dx + \gamma \int_{-\infty}^{\infty} \delta''(x) e^{ikx} dx$

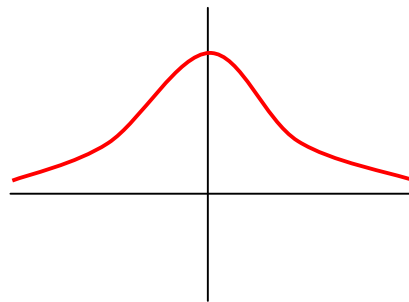
$$\tilde{f}(k) = \alpha - \beta ik + \gamma (-ik)^2 = \alpha - \beta ik - \gamma k^2$$

(b) $\tilde{f}(k) = \int_{-\infty}^{\infty} \frac{1}{x^2 + (\sqrt{2})^2} e^{ikx} dx$

Solving by using residue theorem:

$$= \frac{1}{(x + \sqrt{2}i)(x - \sqrt{2}i)} e^{ikx} \Rightarrow x = \sqrt{2}i$$

$$= \frac{e^{ik\sqrt{2}i}}{2\sqrt{2}i} \times 2\pi i = \frac{e^{-k\sqrt{2}}}{\sqrt{2}} \times \pi = \frac{\pi}{\sqrt{2}} e^{-|k|\sqrt{2}}$$



(c) $\tilde{f}(k) = \int_{-\infty}^0 e^x e^{ikx} dx + \int_0^{\infty} e^{-x} e^{ikx} dx$

$$= \int_{-\infty}^0 e^{x(1+ik)} dx + \int_0^{\infty} e^{x(ik-1)} dx = \left[\frac{e^{x(1+ik)}}{1+ik} \right]_{-\infty}^0 + \left[\frac{e^{x(ik-1)}}{ik-1} \right]_0^{\infty}$$

$$\tilde{f}(k) = \frac{-2}{1+k^2}$$

$$\begin{aligned} \text{(d) } \tilde{f}(k) &= \int_{-\infty}^0 e^{ax} e^{-ikx} \sin bxdx + \int_0^{\infty} e^{-ax} e^{-ikx} \sin bxdx \\ &= \int_{-\infty}^0 e^{x(a-ik)} \sin bxdx + \int_0^{\infty} e^{x(-a-ik)} \sin bxdx \\ &= \left(\frac{e^{x(a-ik)}}{(a-ik)^2 + b^2} [(a-ik) \sin bx - b \cos bx] \right)_{-\infty}^0 + \left(\frac{e^{x(-a-ik)}}{(-a-ik)^2 + b^2} [(-a-ik) \sin bx - b \cos bx] \right)_{0}^{\infty} \\ &= b \left[\frac{1}{(-a-ik)^2 + b^2} - \frac{1}{(a-ik)^2 + b^2} \right] \\ &= b \left[\frac{1}{a^2 + i^2 k^2 + 2aik + b^2} - \frac{1}{a^2 + i^2 k^2 - 2aik + b^2} \right] \\ \tilde{f}(k) &= \frac{-4aib}{(a^2 - k^2 + b^2)^2 + 4a^2 k^2} \end{aligned}$$

$$\begin{aligned} \text{(e) } \tilde{f}(k) &= \int_{-\infty}^{\infty} e^{-a|x|} e^{-ikx} \cos bxdx \\ &= \int_{-\infty}^0 e^{x(a-ik)} \cos bxdx + \int_0^{\infty} e^{-x(a+ik)} \cos bxdx \\ &= \left(\frac{e^{x(a-ik)}}{(a-ik)^2 + b^2} [(a-ik) \cos bx - b \sin bx] \right)_{-\infty}^0 + \left(\frac{e^{-x(a+ik)}}{(-a-ik)^2 + b^2} [-(a+ik) \cos bx - b \sin bx] \right)_{0}^{\infty} \\ &= \left(\frac{a-ik}{(a-ik)^2 + b^2} \right) + \left(\frac{a+ik}{(a+ik)^2 + b^2} \right) \\ &= a \left[\frac{1}{a^2 - k^2 - 2aik + b^2} + \frac{1}{a^2 - k^2 + 2aik + b^2} \right] - ik \left[\frac{1}{a^2 - k^2 - 2aik + b^2} - \frac{1}{a^2 - k^2 + 2aik + b^2} \right] \\ &= a \left[\frac{2a^2 - 2k^2 + 2b^2}{(a^2 - k^2 + b^2)^2 + 4a^2 k^2} \right] - ik \left[\frac{4aik}{(a^2 - k^2 + b^2)^2 + 4a^2 k^2} \right] \end{aligned}$$

$$\tilde{f}(k) = \frac{2a[a^2 - k^2 + b^2] + 4ak^2}{(a^2 + b^2 - k^2)^2 + 4a^2k^2}$$

$$\begin{aligned} \text{(f) } \tilde{f}(k) &= \frac{1}{2a} \int_{-\infty}^{\infty} e^{-ikx} [u(x+a) - u(x-a)] dx \\ &= \frac{1}{2a} (e^{-ik(-a)} - e^{-ika}) = \frac{i}{2a} \left[\frac{e^{ika} - e^{-ika}}{i} \right] \\ \tilde{f}(k) &= \frac{ki \sin ka}{a \quad ka} \quad \left[\frac{\sin \theta}{\theta} \right] \end{aligned}$$

Example: $h(t) = \beta e^{-at} \quad t > 0$

$= 0 \quad \text{otherwise}$

Then find its Fourier transform.

Solution: $H(f) = \int_0^{\infty} \beta e^{-at} e^{-j2\pi ft} dt = \beta \int_0^{\infty} e^{-t(a+j2\pi f)} dt$

$$= \frac{-\beta}{a+j2\pi f} \left[e^{-t(a+j2\pi f)} \right]_0^{\infty} = \frac{\beta}{a+j2\pi f} \left(\frac{\alpha - j2\pi f}{\alpha - j2\pi f} \right)$$

$$H(f) = \frac{\beta\alpha}{\alpha^2 + 4\pi^2 f^2} - j \frac{2\pi f \beta}{\alpha^2 + 4\pi^2 f^2}$$

Example: $H(f) = \frac{\beta\alpha}{\alpha^2 + 4\pi^2 f^2} - j \frac{2\pi f \beta}{\alpha^2 + 4\pi^2 f^2}$

Then find its inverse Fourier transform.

Solution: $h(t) = \int_{-\infty}^{\infty} \frac{\beta\alpha}{\alpha^2 + 4\pi^2 f^2} \cdot e^{j2\pi ft} df - j2\pi\beta \int_{-\infty}^{\infty} \frac{f}{\alpha^2 + 4\pi^2 f^2} e^{j2\pi ft} df$

$$h(t) = \beta\alpha \int_{-\infty}^{\infty} \left(\frac{\cos(2\pi ft)}{\alpha^2 + 4\pi^2 f^2} + \frac{j \sin(2\pi ft)}{\alpha^2 + 4\pi^2 f^2} \right) df - j2\pi\beta$$

$$= \int_{-\infty}^{\infty} \frac{f \cos 2\pi ft}{\alpha^2 + 4\pi^2 f^2} + \frac{fj \sin 2\pi ft}{\alpha^2 + 4\pi^2 f^2} df = \int_{-\infty}^{\infty} \frac{\beta\alpha \cos 2\pi ft}{\alpha^2 + 4\pi^2 f^2} df + j2\pi\beta \int_{-\infty}^{\infty} \frac{f \sin 2\pi ft}{\alpha^2 + 4\pi^2 f^2} df$$

$$= \frac{\beta\alpha}{4\pi^2} \int_{-\infty}^{\infty} \frac{\cos 2\pi ft}{\left(\frac{\alpha}{2\pi}\right)^2 + f^2} df + \frac{2\pi\beta}{4\pi^2} \int_{-\infty}^{\infty} \frac{f \sin 2\pi ft}{\left(\frac{\alpha}{2\pi}\right)^2 + f^2} df$$

$$= \frac{\beta\alpha}{4\pi^2} \frac{\pi}{\left(\frac{\alpha}{2\pi}\right)} e^{-2\pi i \left(\frac{\alpha}{2\pi}\right)} + \frac{2\pi\beta}{4\pi^2} \cdot \pi e^{-2\pi i \left(\frac{\alpha}{2\pi}\right)} = \frac{\beta\alpha}{4\pi^2} \times \frac{\pi \times 2\pi}{\alpha} e^{-i\alpha} + \frac{2\pi\beta}{4\pi^2} \pi e^{-i\alpha}$$

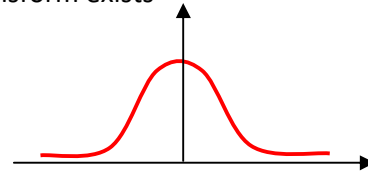
$$h(t) = \frac{\beta}{2} e^{-i\alpha} + \frac{\beta}{2} t^{-i\alpha} = \beta e^{-i\alpha}$$

$$\left(\text{by } \int_{-\infty}^{\infty} \frac{\cos ax}{b^2 + x^2} dx = \frac{\pi}{b} e^{-ab} \quad \& \quad \int_{-\infty}^{\infty} \frac{x \sin ax}{b^2 + x^2} dx = \pi e^{-ab} \right)$$

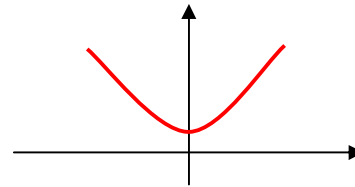
Existence of a Fourier transformation:

Consider a function $h(t)$ if $\int_{-\infty}^{\infty} h(t) dt < \infty$, then Fourier transform exists

- Mod is taken since in general $h(t) + c$
- We can write Fourier transform for $h(t) = e^{-t^2}$



- But we can't write Fourier transform for $h(t) = e^{t^2}$



$\Rightarrow$ So the condition for Fourier integral is $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

$\Rightarrow$ Fourier Transform of Gaussian function is again a Gaussian function.

Example: Gaussian function: $h(t) = e^{-t^2/2\sigma^2} \quad -\infty < t < \infty$

So its Fourier transform will be $H(f) = \int_{-\infty}^{\infty} e^{-t^2/2\sigma^2} e^{-j2\pi ft} dt$

$$H(f) = \sqrt{\frac{\pi}{(1/2\sigma^2)}} e^{-2\pi^2\sigma^2 f^2} = \sqrt{2\pi\sigma^2} e^{-2\pi^2\sigma^2 f^2}$$

$$e^{-t^2/2\sigma^2} \Leftrightarrow \sqrt{2\pi\sigma} e^{-2\pi^2\sigma^2 f^2}$$

Both have the same graph with different heights and areas.

